BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Interfaces with specific hardware.

Q2. Supports separate user sessions and permissions.

Q3. Platform firmware and boot-loading code.

Q4. $B \wedge (\neg A \vee A) = B \wedge 1 = B$.

You Do pages · Independent questions

Q1. Initialises platform [1], loads OS [1], accesses device [1], hosts/manages VMs [1].

Q2. App invokes OS service [1], OS queues request [1], driver translates to device instructions [2], incompatible interface may fail [2].

Q3. Isolation [2], snapshot/reset [1], multiple OSs [1], shared hardware/performance overhead [1], isolation/security not absolute [1].

Exit Ticket · Model answer

False. It shares physical resources through a hypervisor.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

Do Now — Recall previous learning | I Do — Read new knowledge | We Do — Practise with guidance | You Do — Work independently | **Review & Improve** — Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 270 · BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does an OS driver do?

Q2. Define multi-user OS.

Q3. What executes before the OS is loaded?

Q4. Simplify $(\neg A \wedge B) \vee (A \wedge B)$. Show factoring.

Do Now — Recall previous learning | I Do — Read new knowledge | We Do — Practise with guidance | You Do — Work independently | Review & Improve — Check understanding

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.

New learning

Understand — BIOS/UEFI firmware initialises key hardware and starts the boot process.

Trace — Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

Evaluate — A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

Worked example

A printer is connected but software cannot print; name an OS dependency.

1. Predict first — A missing or incompatible driver can prevent correct communication with a device.

2. Model and check — Explain why each value follows from the earlier step.

We Do · Practise with guidance

A teacher needs to test a new OS without replacing their existing install. Choose a solution.

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Host OS + hypervisor + guest VM.
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Predict first

A VM provides a reusable guest environment with snapshots, although it consumes host resources.

Model and check

Explain why each value follows from the earlier step.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

Question 2

Explain how a compatible printer driver enables printing. [6]

Question 3

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

Exit Ticket · Answer before checking

A virtual machine has its own wholly separate physical processor.